

ISutra REST API Reference & Integration Guide

This guide details the complete REST API interface for **ISutra**. It provides procurement software developers, e-Procurement platforms (e.g., GeM, state tender portals), and enterprise systems with full technical documentation on endpoints, request parameters, response schemas, and integration patterns.

 Base URL & Protocol

Environment	Base URL	Notes
Local Development	http://localhost:3001/api	Express server running locally
Production / Vercel	https://<your-deployment-domain>/api	Serverless backend endpoints

All requests and responses use **JSON** format (MIME type: `application/json`), except multipart document uploads which use `multipart/form-data`.

API Endpoints Summary

ISutra REST API

└─ /api/analyze	[POST] Raw text requirement extraction
└─ /api/analysis	
└─ /document	[POST] PDF/DOCX file upload & parsing
└─ /history	[GET] Past analysis sessions
└─ /:id	
└─ (root)	[GET] Analysis details & parameters
└─ /requirements	[PUT] Human-in-the-loop updates
└─ /recommendations	[GET/POST] Deterministic BIS matching
└─ /recommendations/:stdId/gap-analysis	[GET/POST] 4-state requirement gap
└─ /compare	[GET/POST] 11-dimension standards compare
└─ /report	[GET] Structured audit report (JSON)
└─ /report/html	[GET] Printable evaluation report (HTML)
└─ /api/recommendations	
└─ (root)	[POST] Direct requirements matching
└─ /gap-analysis	[POST] Ad-hoc gap analysis
└─ /compare	[POST] Ad-hoc comparison
└─ /api/standards	
└─ (root)	[GET] Directory search & filtering
└─ /categories	[GET] Taxonomy list
└─ /validate-ingestion	[POST] Strict catalog ingestion schema validator
└─ /:id	
└─ (root)	[GET] Standard details & official link
└─ /related	[GET] Allied & normative references
└─ /lifecycle	[GET] 5-stage lifecycle timeline & gazette
└─ /amendments	[GET] Amendments catalog
└─ /regulatory	[GET] Mandatory QCO, CRS & Hallmarking status

1. Procurement Document & Text Ingestion

1.1 Ingest Raw Text Requirements

Extracts 20+ structured procurement parameters from unstructured text specifications or tender paragraphs.

- **Endpoint:** POST /api/analyze
- **Controller:** [analyze](#)
- **Headers:** Content-Type: application/json

Request Body:

```
{
  "text": "Procurement of 120W Outdoor LED Street Light Luminaires for Smart City expressway. Operating vol
  "inputType": "tender_document",
  "language": "en"
}
```

Parameter	Type	Required	Description
text	string	Yes	Tender specification or product requirement text.
inputType	string	No	'product_description' 'technical_specification' 'tender_document'. Default: 'product_description'.
language	string	No	'en' 'hi' 'te'. Default: detected automatically by script heuristic.

Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "req-1711200000000-abc123",
    "status": "completed",
    "inputType": "tender_document",
    "detectedLanguage": "en",
    "extractedRequirements": {
      "product_name": "Outdoor LED Street Light Luminaire",
      "category": "Luminaires",
      "subcategory": "Roadway & Street Lighting",
      "primary_material": "Die-cast Aluminum",
      "application_domain": "Outdoor Highway / Expressway Lighting",
      "operating_environment": "Outdoor Severe Weather",
      "electrical_ratings": {
        "wattage": "120W",
        "voltage": "240V AC",
        "power_factor": "> 0.95",
        "surge_protection": "10kV"
      },
      "performance_metrics": {
        "luminous_efficacy": ">= 120 lm/W",
        "cct": "5700K"
      },
      "safety_features": [
        "IP66 Ingress Protection",
        "Over-voltage and Surge Protection"
      ],
      "testing_requirements": [
        "Ingress Protection Test",
        "Surge Immunity Test"
      ]
    },
    "provenance": {
      "sourceLength": 328,
      "aiProvider": "gemini-2.5-flash",
      "extractedAt": "2026-10-03T18:24:00.000Z",
      "isOcrDerived": false
    }
  }
}
```

1.2 Ingest Digital PDF / DOCX Document

Ingests tender files, extracts text streams, validates magic-byte headers, and extracts requirements.

- **Endpoint:** `POST /api/analysis/document` (or `/api/analysis/upload`)
- **Service:** `documentExtractionService.ts`
- **Headers:** `Content-Type: multipart/form-data`

Multipart Form Parameters:

- `document`: File binary (supported extensions: `.pdf`, `.docx`, `.doc`). Maximum size: 15 MB.
- `inputType`: (optional) `'tender_document'`.
- `language`: (optional) `'en'` | `'hi'` | `'te'`.

Response (`200 OK`):

Same format as `POST /api/analyze`, with `file_name`, `pageCount`, and paragraph statistics in provenance.

Error Responses:

- `400 Bad Request`: File missing, or unsupported extension (e.g. `.exe`, `.zip`).
- `422 Unprocessable Entity`: Scanned PDF detected without readable text stream:

```
{
  "success": false,
  "error": "Scanned document detected. No digital text stream could be extracted.",
  "code": "SCANNED_DOCUMENT_NO_TEXT",
  "remediation": "Upload a digital PDF generated directly from Word/CAD, or enable OCR processing in dep
```

2. Human-in-the-Loop Requirement Review

Public procurement cannot legally rely on unchecked AI perceptions. Officers review, modify, and confirm parameters before matching.

2.1 Fetch Analysis Session

- **Endpoint:** `GET /api/analysis/:id`
- **Response (`200 OK`):** Returns current extraction state, verification status, and raw extracted parameter tree.

2.2 Update & Confirm Requirements

- **Endpoint:** `PUT /api/analysis/:id/requirements`
- **Controller:** `updateRequirements`
- **Headers:** `Content-Type: application/json`

Request Body:

```
{
  "confirmedRequirements": {
    "product_name": "Outdoor LED Street Light Luminaire",
    "category": "Luminaires",
    "application_domain": "Highway / Roadway",
    "electrical_ratings": {
      "wattage": "150W",
      "surge_protection": "10kV"
    },
    "safety_features": ["IP66 Ingress Protection"]
  },
  "reviewNotes": "Upgraded wattage requirement from 120W to 150W as per revised municipal budget.",
  "isConfirmed": true
}
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "req-1711200000000-abc123",
    "isConfirmed": true,
    "confirmedAt": "2026-10-03T18:25:30.000Z",
    "confirmedRequirements": { ... }
  }
}
```

3. Deterministic Standards Recommendation Engine

3.1 Get Recommendations for Analysis Session

Evaluates confirmed parameters against verified BIS standards using the 6-factor deterministic scoring engine.

- **Endpoint:** GET /api/analysis/:id/recommendations (or POST)
- **Service:** [standardsMatcher.ts](#)

Response (200 OK):

```
{
  "success": true,
  "data": {
    "analysisId": "req-1711200000000-abc123",
    "totalEvaluated": 40,
    "recommendations": [
      {
        "rank": 1,
        "standardId": "is-10322-5-3",
        "standardNumber": "IS 10322 (Part 5/Sec 3): 2012",
        "title": "Luminaires - Particular Requirements - Luminaires for Road and Street Lighting",
        "category": "Luminaires",
        "relevanceScore": 0.825,
        "categoryClassification": "high",
        "categoryLabel": "HIGH RELEVANCE",
        "specificityTier": 1,
        "specificityLabel": "Tier 1: Exact Product & Application Match",
        "officialSourceUrl": "https://services.bis.gov.in/php/BIS_2.0/bisconnect/knownyourstandards/indian_s",
        "factorBreakdown": {
          "productCategory": {
            "status": "matched",
            "score": 1.0,
            "weight": 0.30,
            "contribution": 0.300,
            "evidence": ["Exact category match: Luminaires"]
          },
          "keywordsTitleScope": {
            "status": "matched",
            "score": 0.85,
            "weight": 0.25,
            "contribution": 0.2125,
            "evidence": ["Title aligns with 'Road and Street Lighting'", "Scope covers public thoroughfares"]
          },
          "applicationDomain": {
            "status": "matched",
            "score": 1.0,
            "weight": 0.15,
            "contribution": 0.150,
            "evidence": ["Matched application: Highway / Roadway Lighting"]
          },
          "environment": {
            "status": "matched",
            "score": 0.90,
            "weight": 0.10,
            "contribution": 0.090,
            "evidence": ["Outdoor severe weather and dust resistance covered in clause 4.2"]
          },
          "technicalParameters": {
            "status": "matched",
            "score": 0.36,
            "weight": 0.10,
            "contribution": 0.036,
            "evidence": ["Standard defines mechanical/electrical limits; lamp performance governed by IS 16"]
          },
          "safetyTesting": {
            "status": "matched",
```

```

        "score": 0.37,
        "weight": 0.10,
        "contribution": 0.037,
        "evidence": ["Ingress test IP66 cross-references IS/IEC 60529"]
    }
},
"auditChain": [
    { "stage": "user_input", "detail": "Outdoor LED Street Light Luminaire for expressway" },
    { "stage": "extracted_requirement", "detail": "Category: Luminaires, Application: Roadway" },
    { "stage": "matching_signal", "detail": "Product (30%) + Title/Scope (25%) + Application (15%)" },
    { "stage": "standard_data", "detail": "IS 10322 (Part 5/Sec 3): 2012 Clause 1.1 Scope" },
    { "stage": "official_bis_source", "detail": "https://services.bis.gov.in" }
]
}
]
}
}

```

4. Gap Analysis & Standards Comparison

4.1 Requirement Gap Analysis

Audits tender specifications against the selected standard's scope, classifying each requirement into 4 non-hallucinatory reference coverage states.

- **Endpoint:** `GET /api/analysis/:id/recommendations/:standardId/gap-analysis`
- **Service:** `requirementGapAnalyzer.ts`

Response (200 OK):

```
{
  "success": true,
  "data": {
    "standardNumber": "IS 10322 (Part 5/Sec 3): 2012",
    "standardTitle": "Luminaires - Road and Street Lighting",
    "coverageNotice": "Reference Coverage Audit. This analysis reflects textual scope coverage in verified",
    "summary": {
      "supportedCount": 5,
      "notSupportedCount": 0,
      "notAvailableCount": 2,
      "needsVerificationCount": 1
    },
    "parameters": [
      {
        "name": "Ingress Protection (IP66)",
        "status": "supported",
        "coverageBadge": "SUPPORTED IN STANDARD",
        "clauseReference": "Clause 4.3 (referencing IS/IEC 60529)",
        "notes": "Clause mandates minimum IP43 for optical chamber; IP66 tested per Section 4."
      },
      {
        "name": "Luminous Efficacy (120 lm/W)",
        "status": "not_available",
        "coverageBadge": "NOT GOVERNED BY THIS CODE",
        "clauseReference": "Out of Scope",
        "notes": "Safety standard IS 10322 does not prescribe photometric efficacy. Refer to allied standar"
      },
      {
        "name": "Surge Protection (10kV)",
        "status": "needs_verification",
        "coverageBadge": "REQUIRES MANUAL VERIFICATION",
        "clauseReference": "Clause 11.2 (Surge Immunity)",
        "notes": "Standard specifies basic surge limits. 10kV high-voltage municipal surge requires externa"
      }
    ]
  }
}
```

4.2 Side-by-Side Standards Comparison

Compares 2 or 3 standards across 11 neutral dimensions without bias or subjective winner declarations.

- **Endpoint:** `POST /api/analysis/:id/compare`
- **Service:** `standardsComparator.ts`

Request Body:

```
{
  "standardIds": ["is-10322-5-3", "is-16107-2-1"]
}
```

Response (`200 OK`):

Returns an 11-dimension comparison matrix covering:

1. Primary Scope
 2. Product Type
 3. Applicable Domain
 4. Electrical & Safety Focus
 5. Mandatory Test Methods
 6. Environmental Ingress Specs
 7. Target Users
 8. Complementary Relationship
 9. Reaffirmation Date
 10. Regulatory Mandate (QCO)
 11. Gazette Source Reference
-

5. Standards Directory, Lifecycle & Regulatory

5.1 Search & Filter Standards

- **Endpoint:** `GET /api/standards`
- **Controller:** `standardsController.ts`
- **Query Parameters:**
 - `search` : Keyword or standard number (e.g. `10322` , `cable` , `cement`).
 - `category` : Category name (e.g. `Luminaires` , `Civil Construction` , `Power Cables`).
 - `status` : `'active'` | `'under_revision'` | `'withdrawn'` .
 - `page` : Page number (default: 1).
 - `limit` : Items per page (default: 20).

5.2 Allied & Normative Standards

- **Endpoint:** `GET /api/standards/:id/related`
- **Service:** `standardsRelationshipService.ts`

Categorizes relationships into:

- `normative_reference` : Mandatory standard cited in normative text.
- `test_method` : Standard specifying physical/electrical testing procedures.
- `safety_standard` : General electrical or fire safety code.
- `installation_standard` : Guidelines for mounting and civil execution.

5.3 Lifecycle & Amendment History

- **Endpoint:** GET /api/standards/:id/lifecycle
- **Service:** [standardLifecycleService.ts](#)

Returns the 5-stage lifecycle history: BASE STANDARD → REVISION → AMENDMENTS → REAFFIRMATION → VERIFIED BIS PORTAL .

5.4 Regulatory & QCO Status

- **Endpoint:** GET /api/standards/:id/regulatory
- **Service:** [regulatoryService.ts](#)

Returns 1 of 3 strict regulatory states:

1. VERIFIED APPLICABILITY : Mandatory QCO gazette notified (e.g. Steel Quality Control Order for IS 1786).
2. VERIFICATION REQUIRED : Transitional enforcement or recent Ministry draft.
3. NO VERIFIED REGULATORY RECORD : Standard is voluntary or no gazette QCO exists.

6. Procurement Intelligence Reports

6.1 Generate JSON Procurement Report

- **Endpoint:** GET /api/analysis/:id/report
- **Response:** Full structured audit dossier with confirmed requirements, top recommendations, factor breakdowns, gap analysis, allied codes, and gazette links.

6.2 Export Printable HTML Report

- **Endpoint:** GET /api/analysis/:id/report/html
- **Response:** Pre-styled, audit-grade printable HTML report ready for printing, filing, or PDF export (Ctrl+P).

7. Catalog Ingestion Schema Validator

Validates candidate Indian Standard records before ingestion to ensure zero hallucination, mandatory official sources, and proper typing.

- **Endpoint:** POST /api/standards/validate-ingestion
- **Service:** [standardsIngestionService.ts](#)

Request Body:

```
{
  "id": "is-10322-5-3",
  "standard_number": "IS 10322 (Part 5/Sec 3): 2012",
  "title": "Luminaires - Particular Requirements - Luminaires for Road and Street Lighting",
  "category": "Luminaires",
  "scope": "This standard specifies requirements for luminaires for road, street lighting and other public
  "source_url": "https://services.bis.gov.in/php/BIS_2.0/bisconnect/knownyourstandards/indian_standards/isde
  "last_verified": "2026-03-01T00:00:00.000Z",
  "status": "active"
}
```

Validation Rules Enforced:

- 1. `id` and `standard_number` must match valid IS syntax (`IS \d+...`).
- 2. `source_url` must use secure HTTPS protocol on recognized government or BIS domains (`services.bis.gov.in` , `bis.gov.in` , `egazette.gov.in` , or `crsbis.in`). Non-government URLs are rejected.
- 3. `last_verified` date must be present and parseable as ISO 8601.
- 4. Scope cannot be empty.

8. HTTP Status Codes & Error Handling

HTTP Status	Meaning	Typical Trigger
200 OK	Success	Request succeeded. Response body contains requested payload.
400 Bad Request	Invalid Input	Missing required fields, malformed JSON, or unsupported file type.
404 Not Found	Resource Missing	Non-existent <code>analysisId</code> or <code>standardId</code> .
422 Unprocessable Entity	Extraction Limitation	Scanned PDF document without readable text stream.
500 Internal Server Error	Server Exception	Uncaught runtime error; fallback gracefully engages if database/AI is offline.

All error responses return structured JSON:

```
{  
  "success": false,  
  "error": "Human readable error description",  
  "code": "ERROR_CONSTANT_CODE",  
  "details": null  
}
```